



Molecular SSINet: Applying Self-Supervised Discovering of Interpretable Features to Molecule Generation

Pimpinelli Francesco 2214340
REINFORCEMENT LEARNING A.A 25/26

PHARMACEUTICALS

Novo Nordisk partners with OpenAI as AI drug discovery hopes mount

PUBLISHED TUE, APR 14 2026-2:20 AM EDT | UPDATED TUE, APR 14 2026-8:33 AM EDT

2026: the year AI stops being optional in drug discovery

By  Drug Target Review | January 19, 2026



AI is moving from a supporting role into the core of drug discovery. By 2026, it is expected to shape how targets are chosen, how biology is analysed and how development decisions are made.

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Accurate structure prediction of biomolecular interactions with AlphaFold 3

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The Context: Interpretability

ANTHROPIC

Interpretability

The mission of the Interpretability team is to discover and understand how large language models work internally, as a foundation for AI safety and positive outcomes.

Research teams: [Alignment](#) [Economic Research](#) [Interpretability](#) [Societal Impacts](#)

Interpretability

Tracing the thoughts of a large language model

27 Mar 2025

“Better understanding of the decision-making processes”

Emotion concepts and their function in a large language model

2 Apr 2026



The Context: MoleGuLAR &



SSINet

JCIMJOURNAL OF
CHEMICAL INFORMATION
AND MODELINGpubs.acs.org/jcim

Article

MoleGuLAR: Molecule Generation Using Reinforcement Learning with Alternating Rewards

Manan Goel, Shampa Raghunathan, Siddhartha Laghuvarapu, and U. Deva Priyakumar*



Cite This: *J. Chem. Inf. Model.* 2021, 61, 5815–5826

2712

IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 44, NO. 5, MAY 2022

Self-Supervised Discovering of Interpretable Features for Reinforcement Learning

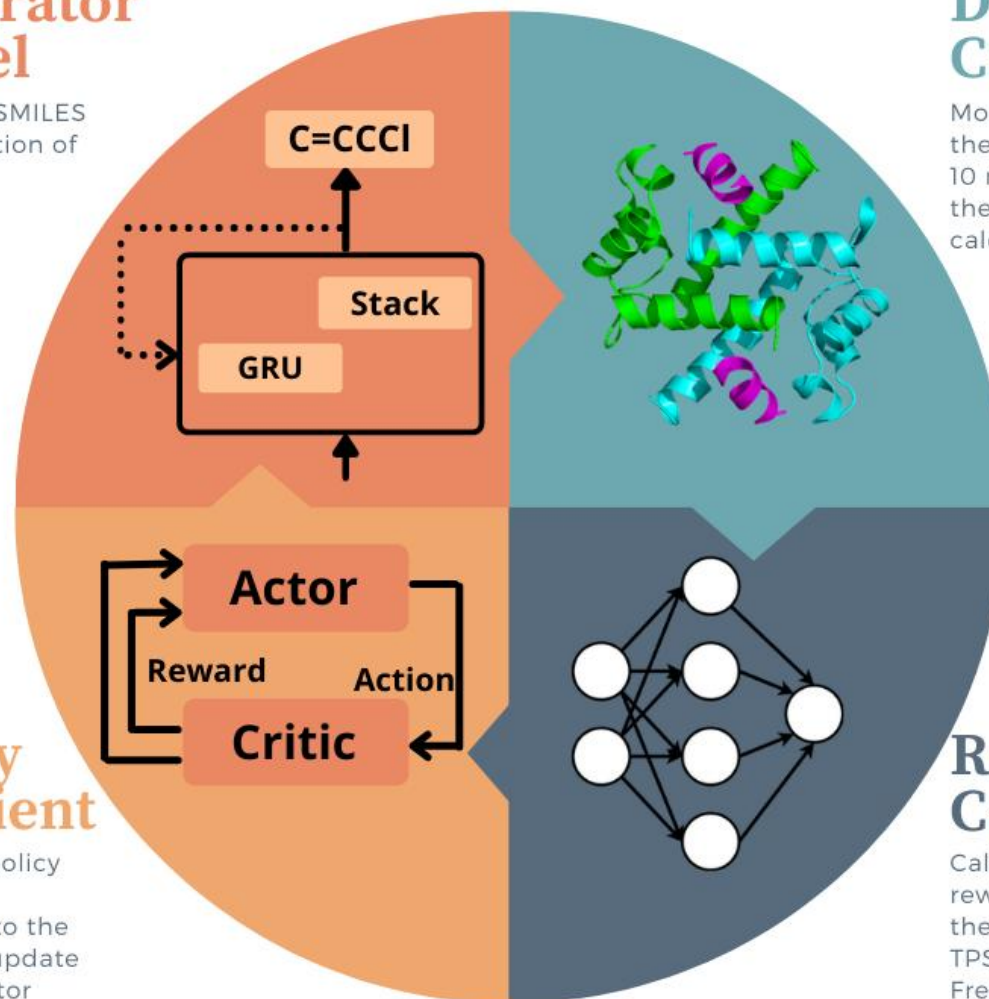
Wenjie Shi¹, Gao Huang², Member, IEEE, Shiji Song³, Senior Member, IEEE,
Zhuoyuan Wang, Tingyu Lin, and Cheng Wu⁴



MoleGuLAR: The Pipeline

Generator Model

Generates SMILES representation of molecule



Docking Calculation

Molecule is docked to the target protein for 10 runs and affinity of the best pose is calculated

Reward Calculation

Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy

Policy Gradient

Calculate policy gradient according to the reward to update the generator parameters



MoleGuLAR: 😊 SMILES

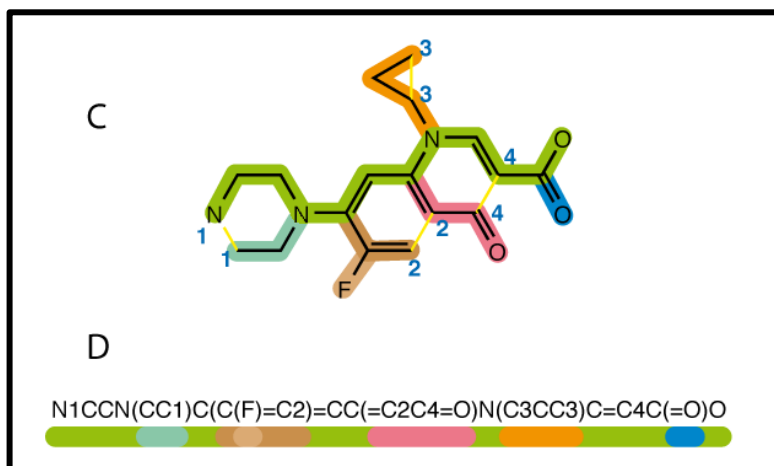
SMILES, a Chemical Language and Information System. 1. Introduction to Methodology and Encoding Rules

DAVID WEININGER

Medicinal Chemistry Project, Pomona College, Claremont, California 91711

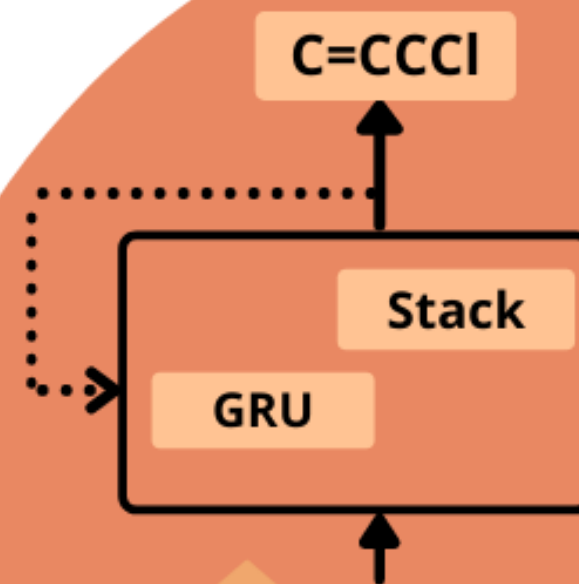
Received June 17, 1987

SMILES (Simplified Molecular Input Line Entry System) is a chemical notation system designed for modern chemical information processing. Based on principles of molecular graph theory, SMILES allows rigorous structure specification by use of a very small and natural grammar. The SMILES notation system is also well suited for high-speed machine processing. The resulting ease of usage by the chemist and machine compatibility allow many highly efficient chemical computer applications to be designed including generation of a unique notation, constant-speed (zeroth order) database retrieval, flexible substructure searching, and property prediction models.



Generator Model

Generates SMILES representation of molecule





MoleGuLAR: Data & Target Proteins



Pre-Trained on
ChEMBL



SARS – CoV – 2 M_{pro}

Protein essential for the viral transcription and replication of the **COVID-19 virus**.

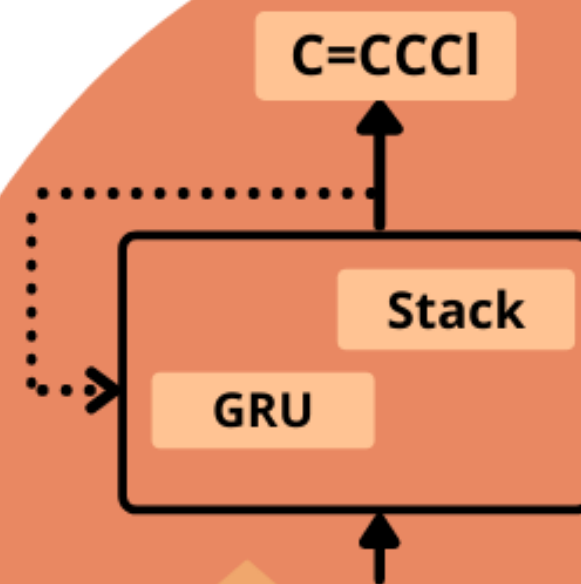


TTBK1

Target protein for treating some **neurodegenerative** diseases, such as Alzheimer's.

Generator Model

Generates SMILES representation of molecule

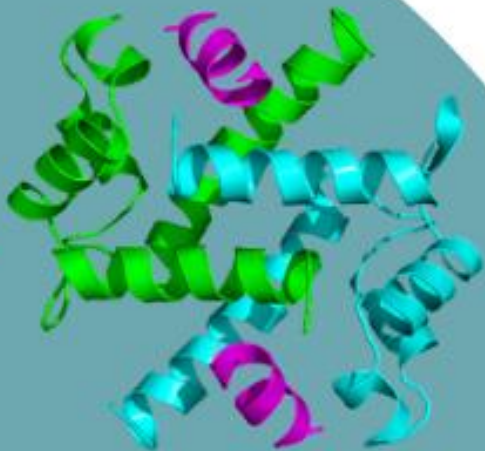




MoleGuLAR: BA Estimation

Docking Calculation

Molecule is docked to the target protein for 10 runs and affinity of the best pose is calculated



Graph Embeddings + Random Forest

MAE:	R²:
0.87	0.55

Mol2Vec + Random Forest

MAE:	R²:
0.47	0.91

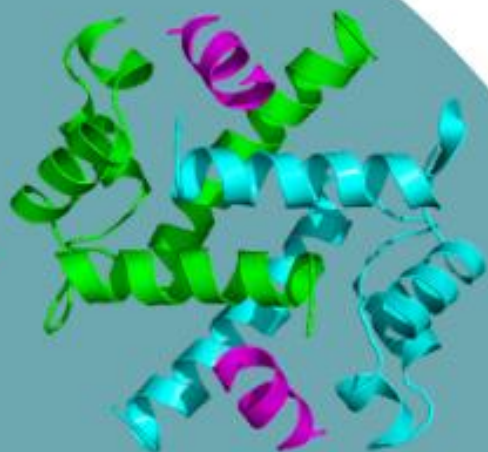
Graph Isomorphism Network (GIN)

MAE:	R²:
0.45	0.93

BA Estimation: Chosen Architecture

Docking Calculation

Molecule is docked to the target protein for 10 runs and affinity of the best pose is calculated



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Graph Isomorphism Network (GIN)

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🤝 BA Estimation: Chosen Architecture

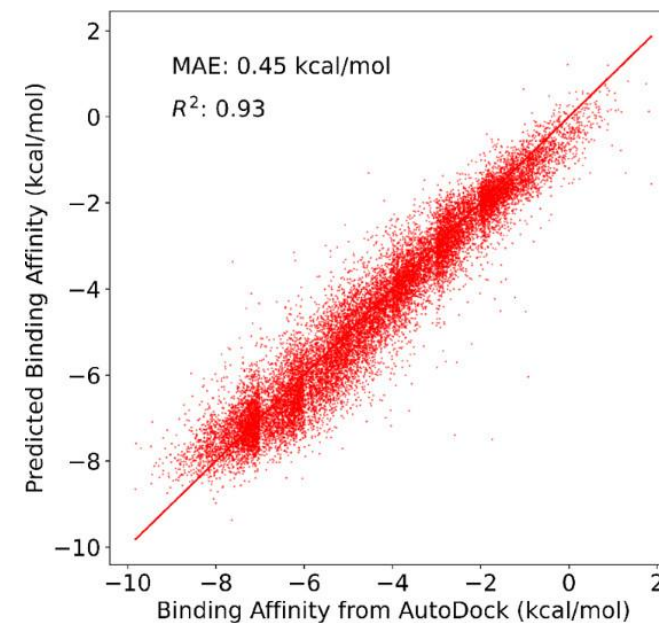
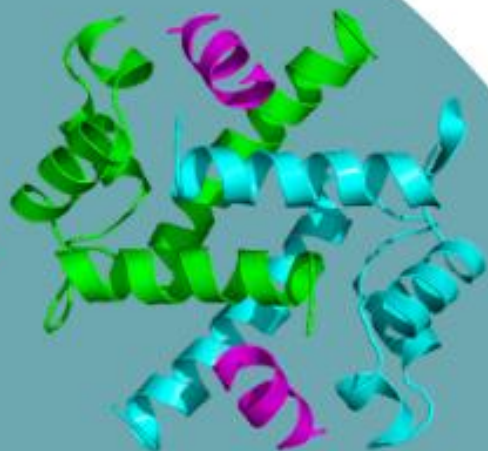
Graph Isomorphism Network (GIN)

MAE:
0.45

R^2 :
0.93

Docking Calculation

Molecule is docked to the target protein for 10 runs and affinity of the best pose is calculated



(a) BA predicted by GIN vs. ground truth values from AutoDock

MoleGuLAR: Desired Properties

Reward Calculation

Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy



TPSA

Surface sum over all **polar atoms** or molecules, including their attached hydrogen atoms



Binding Affinity

Strength of the **non-covalent interaction** between two molecules



LogP

Measures a molecule's **lipophilicity**, how well it dissolves in fats vs water



QED

A 0-to-1 scoring method to map **resemblance** to known oral drugs



Hydration Free Energy

The amount of energy released when one mole of ions undergoes **solvation**

MoleGuLAR: Desired Properties

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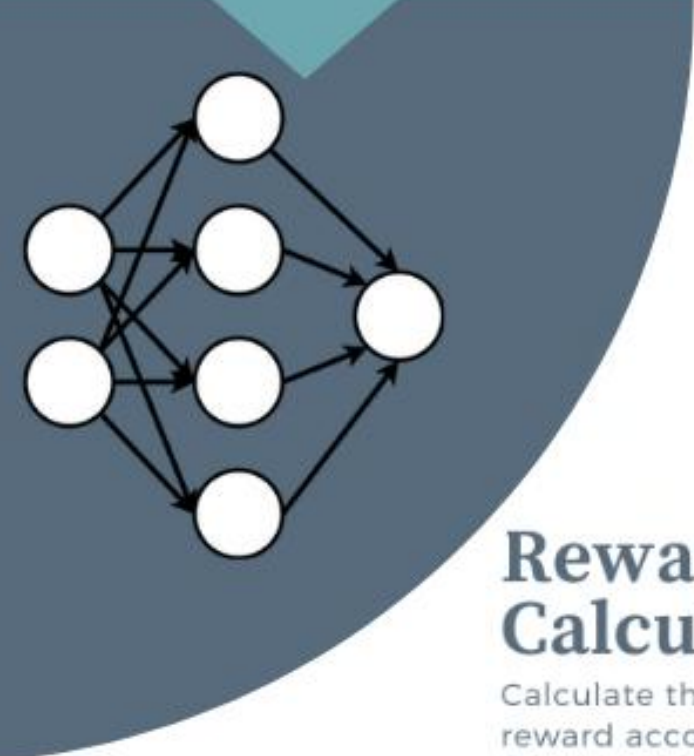
A 0-to-1 scoring method to map **resemblance** to known oral drugs



Hydration Free Energy

The amount of energy released when one mole of ions undergoes **solvation**

MoleGuLAR: 🏆 Reward Strategies



Reward Calculation

Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy



Exponential

Strong push toward low binding energy



Linear

Constant gradient based on the score minus a threshold



Logarithmic

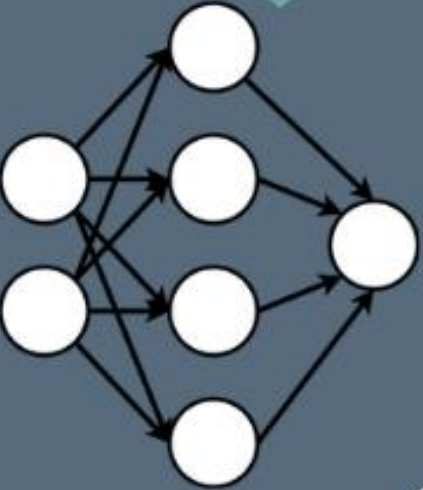
Gentle at first but becomes **aggressive** as scores approach zero



Squared

Heavily punishes low-quality molecules

MoleGuLAR: 🏆 Reward Strategies



Reward Calculation

Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy



Weighted Sum Rewards

Among all the **Desired Properties**, with fixed weights



Alternating Rewards

All weights are set to **0** except for **one** specific property, which is cycled at regular intervals

MoleGuLAR: Reward Strategies

Reward Calculation

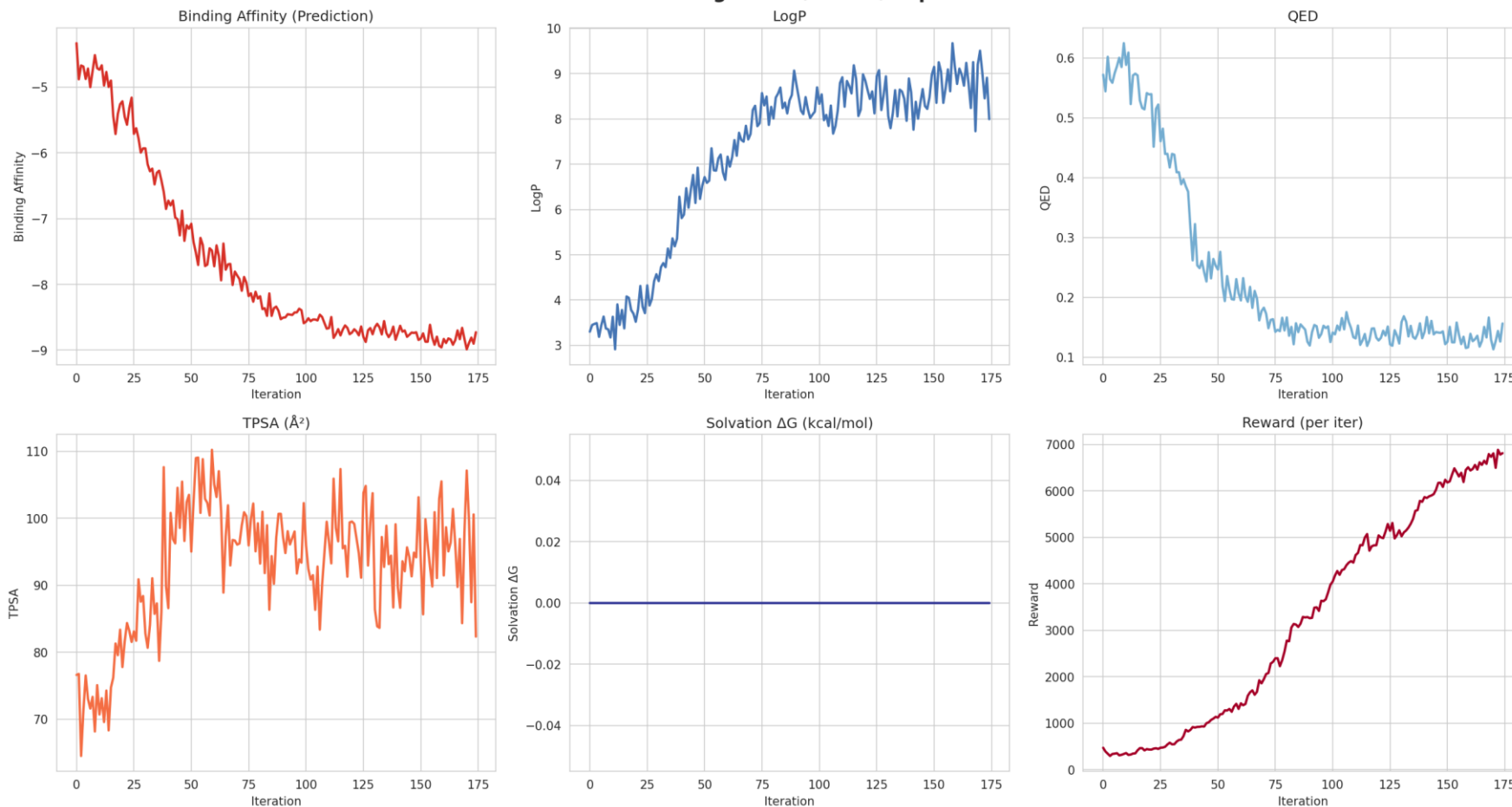
Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy



Weighted Sum Rewards

Among all the **Desired Properties**, with fixed weights

MoleGuLAR Training — GIN / 4BTK / exponential



MoleGuLAR: Reward Strategies

Reward Calculation

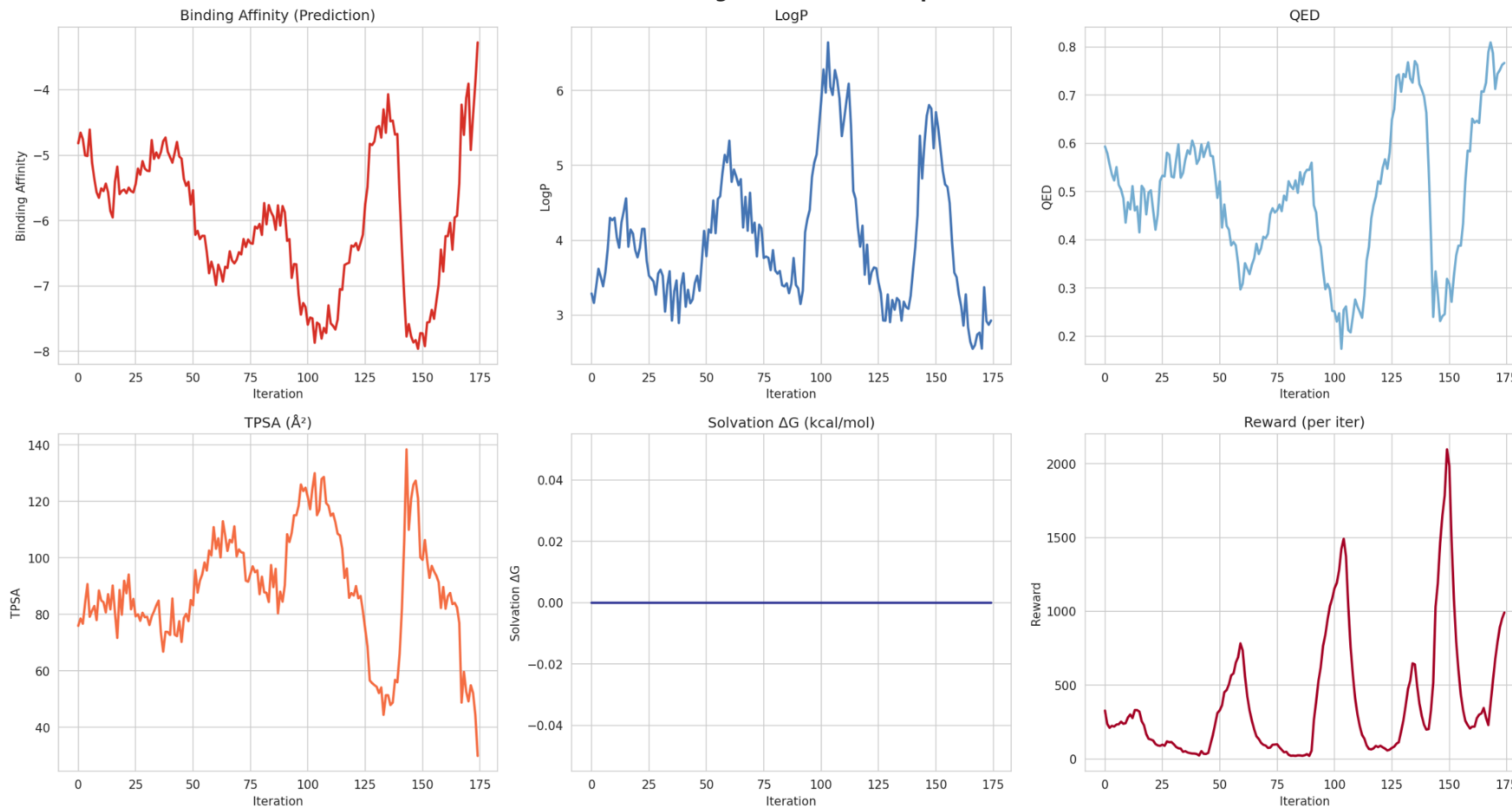
Calculate the total reward according to all the objectives like LogP, TPSA, QED and Hydration Free Energy



Alternating Rewards

All weights are set to **0** except for **one** specific property, which is cycled at regular intervals

MoleGuLAR Training — GIN / 4BTK / exponential





MoleGuLAR: 🤖 The RL Model

N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)C=C4C(=O)O

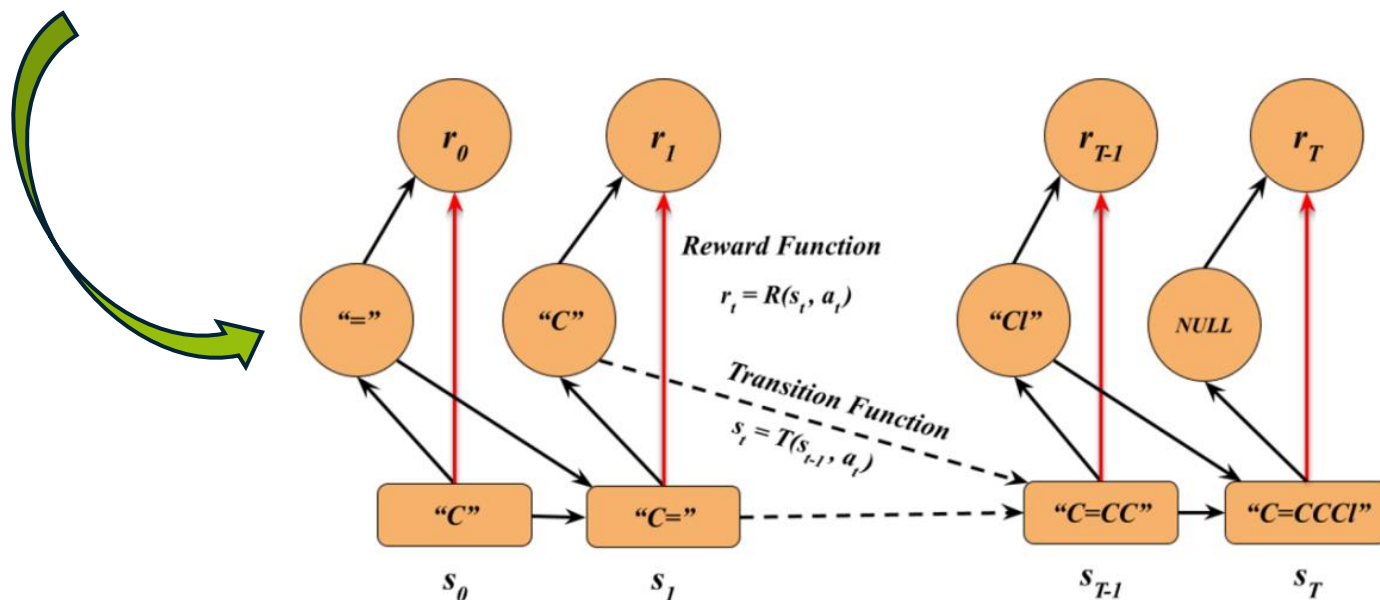


Figure 3. Illustration of SMILES string generation as a Markov decision process. At each state s_t , the agent performs an action a_t to give the updated state s_{t+1} and provide a reward according to the state.

Policy Gradient

Calculate policy gradient according to the reward to update the generator parameters





The RL Model : 🦾 REINFORCE

Expected Reward Function

$$R(\Theta) = \mathbb{E}[r(s_T)|s_0, \Theta] = \sum_{s_T \in \mathcal{S}} p_{\Theta}(s_T) r(s_T)$$

$\mathcal{S}$ has exponential size

$$R(\Theta) = \mathbb{E}[r(s_T)|s_0, \Theta] = \mathbb{E}_{a_1 \sim p_{\Theta}(a_1|s_0)} \dots \mathbb{E}_{a_T \sim p_{\Theta}(a_T|s_{T-1})} r(s_T)$$

Differentiation Rule

$$\delta_{\Theta} f(\Theta) = f(\Theta) \frac{\delta_{\Theta} f(\Theta)}{f(\Theta)} = f(\Theta) \delta_{\Theta} \log f(\Theta)$$

Gradient

$$\delta_{\Theta} R(\Theta) = \mathbb{E}_{a_1 \sim p_{\Theta}(a_1|s_0)} \dots \mathbb{E}_{a_T \sim p_{\Theta}(a_T|s_{T-1})} \left[\sum_{t=1}^T \delta_{\Theta} \log p_{\Theta}(a_t|s_{t-1}) \right] r(s_T)$$

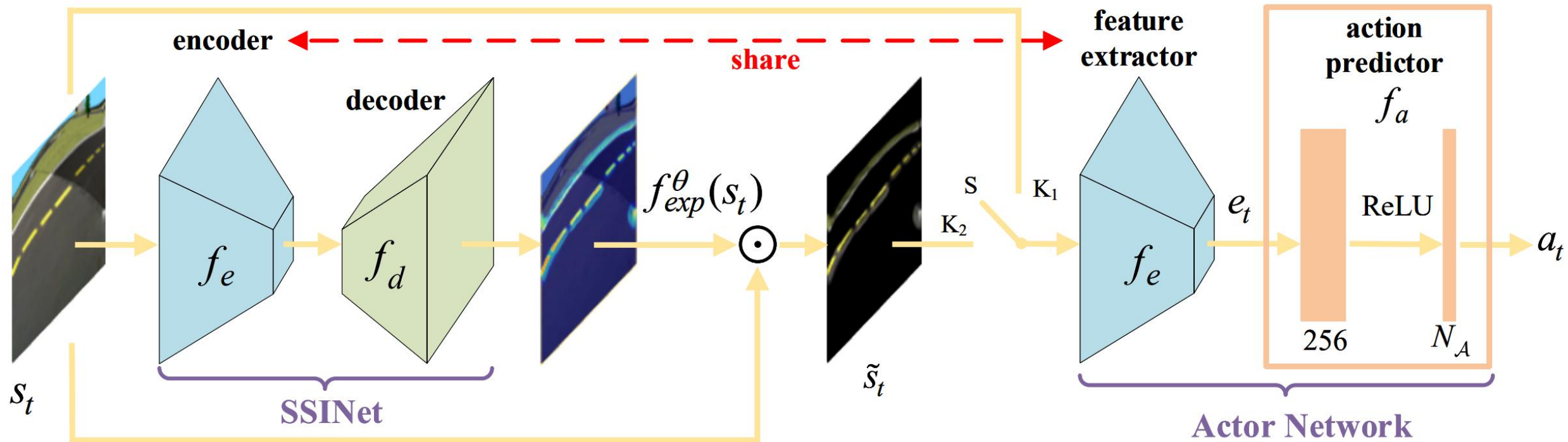
Policy Gradient

Calculate policy gradient according to the reward to update the generator parameters





SSINet: The Architecture

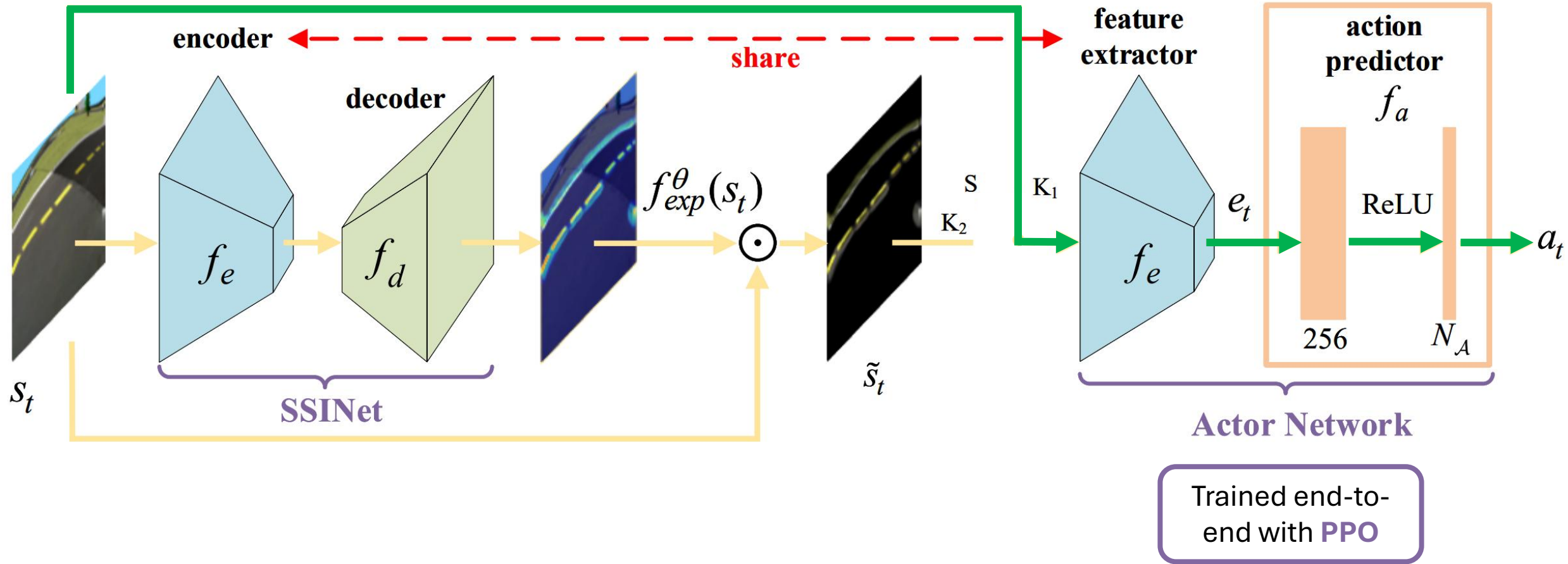




The Training: Stage 1

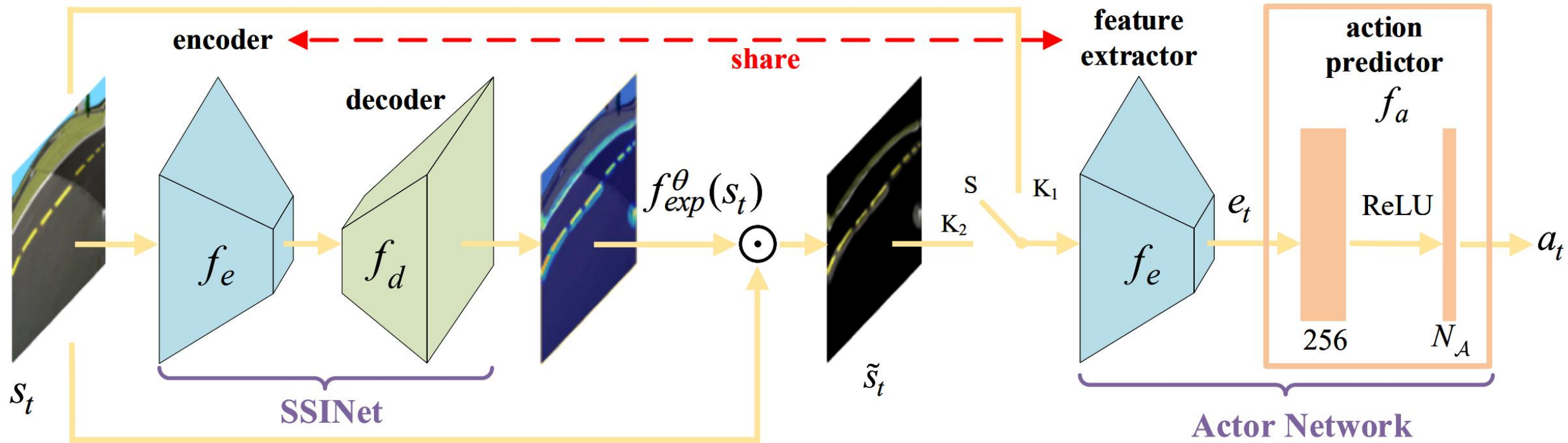


The Actor Network





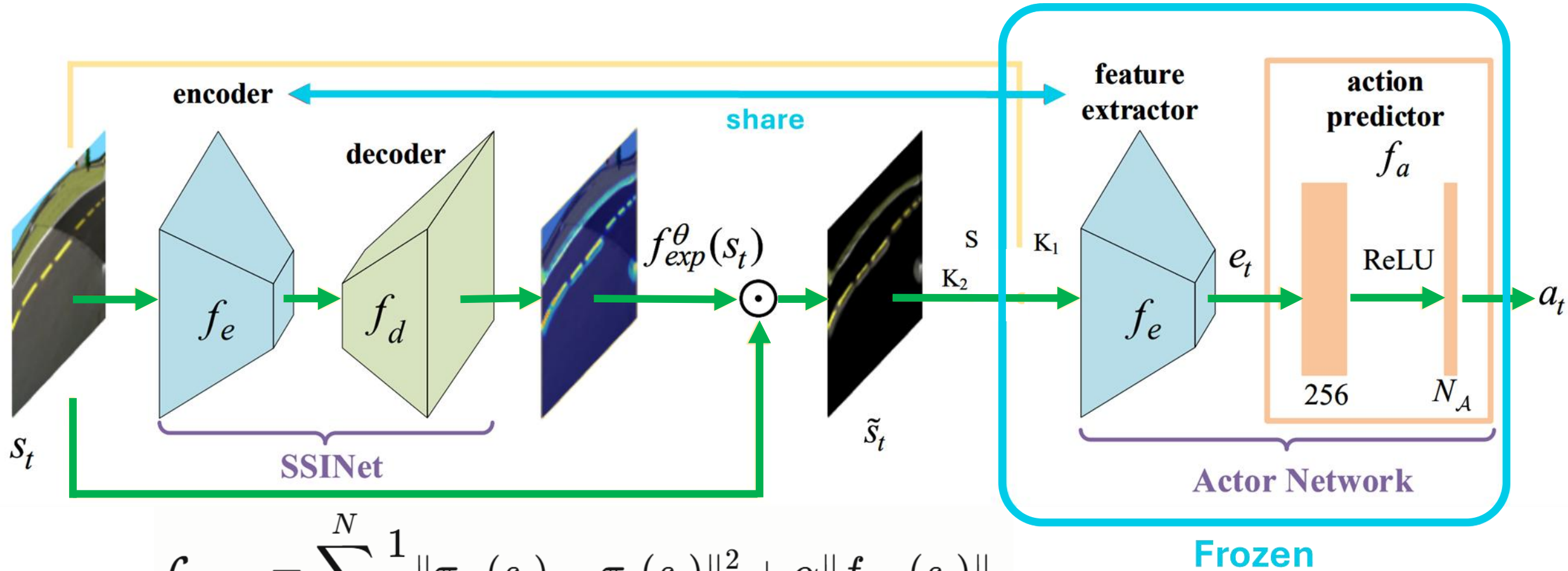
SSINet: The Training (Self-Supervised)



After convergence becomes the **expert policy** to collect the state-action pairs from the environment



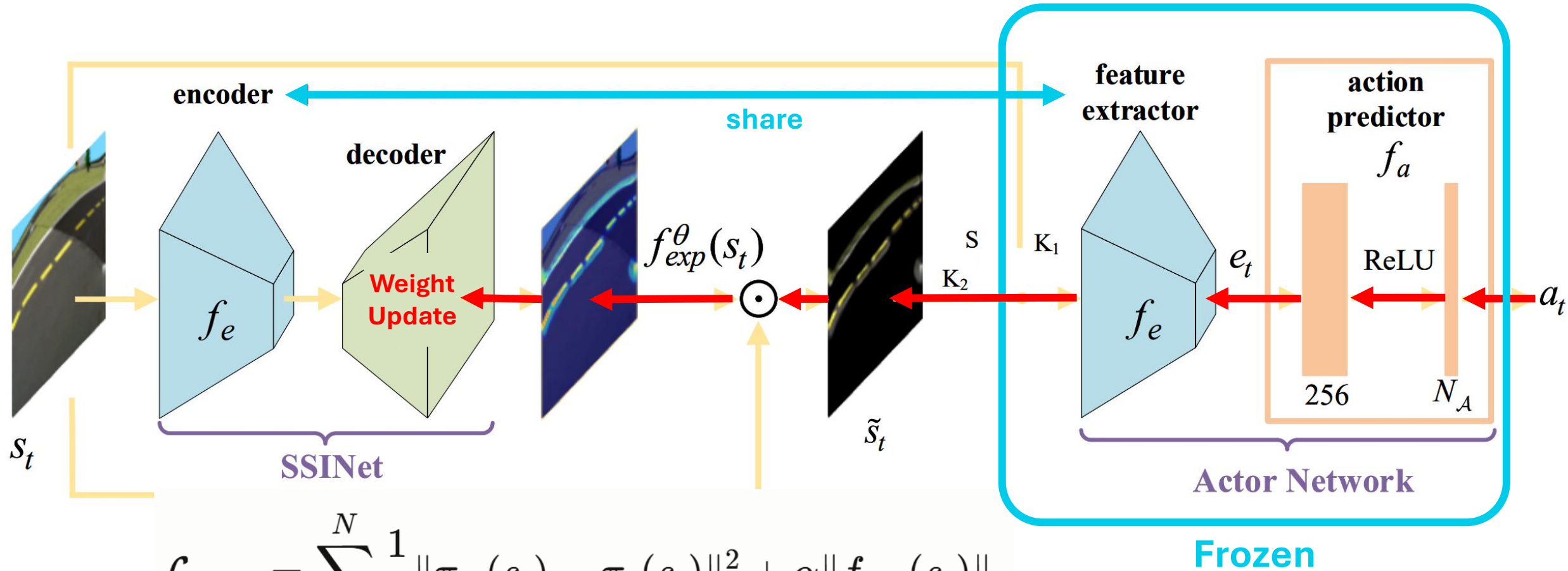
The Training: Stage 2 The SSINet



$$\mathcal{L}_{\text{mask}} = \sum_{i=1}^N \frac{1}{2} \|\pi_m(s_i) - \pi_e(s_i)\|_2^2 + \alpha \|f_{\text{exp}}(s_i)\|_1$$



The Training: Stage 2 The SSINet





SSINet: 2 Two Core Properties

$$\mathcal{L}_{\text{mask}} = \sum_{i=1}^N \frac{1}{2} \|\pi_m(s_i) - \pi_e(s_i)\|_2^2 + \alpha \|f_{\text{exp}}(s_i)\|_1$$



Maximum Behavior Resemblance

The masked state must yield (almost) the **same action** as the original state



Minimum Region Retaining

The mask should retain **as little of the image** as possible.



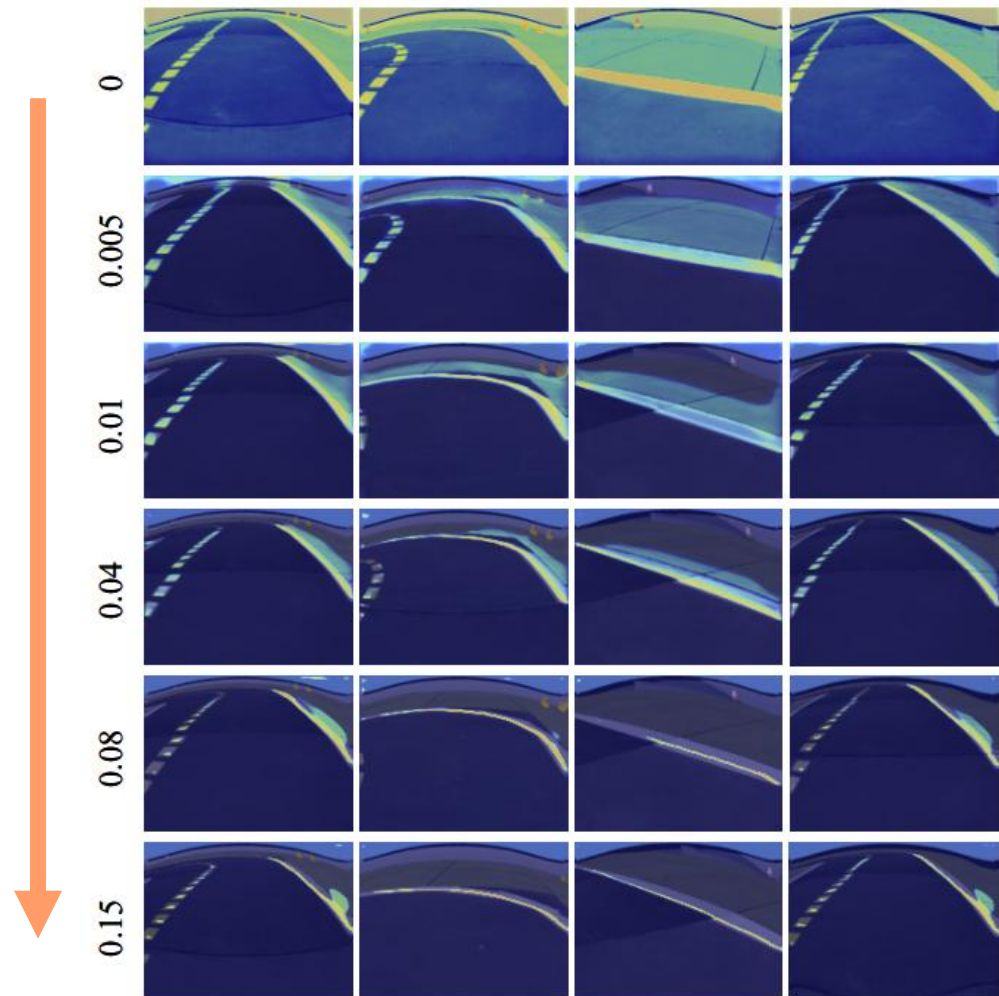
SSINet: 2 Two Core Properties

$$- \|\pi_e(s_i)\|_2^2 + \boxed{\alpha} \|f_{\text{exp}}(s_i)\|_1$$



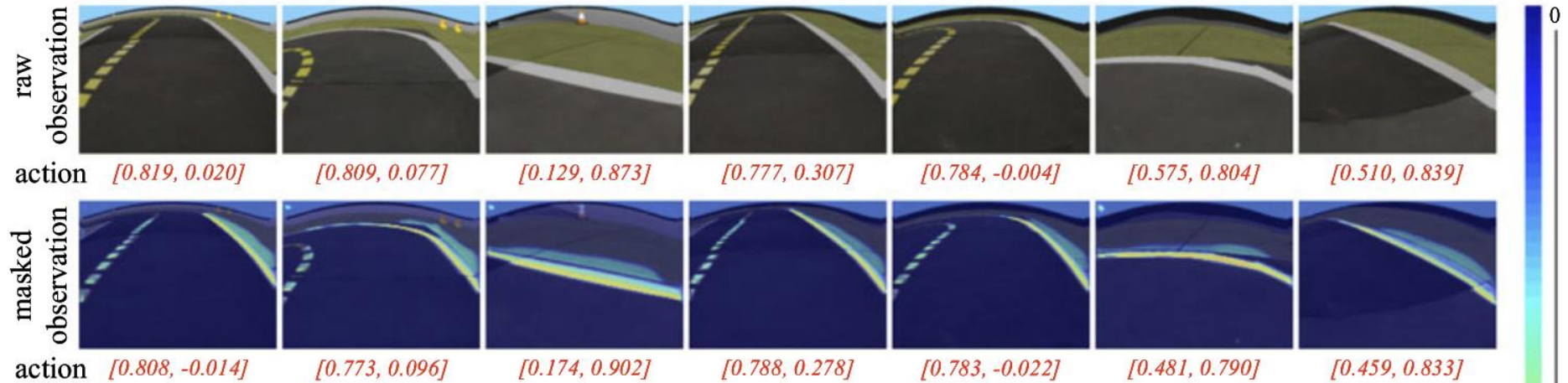
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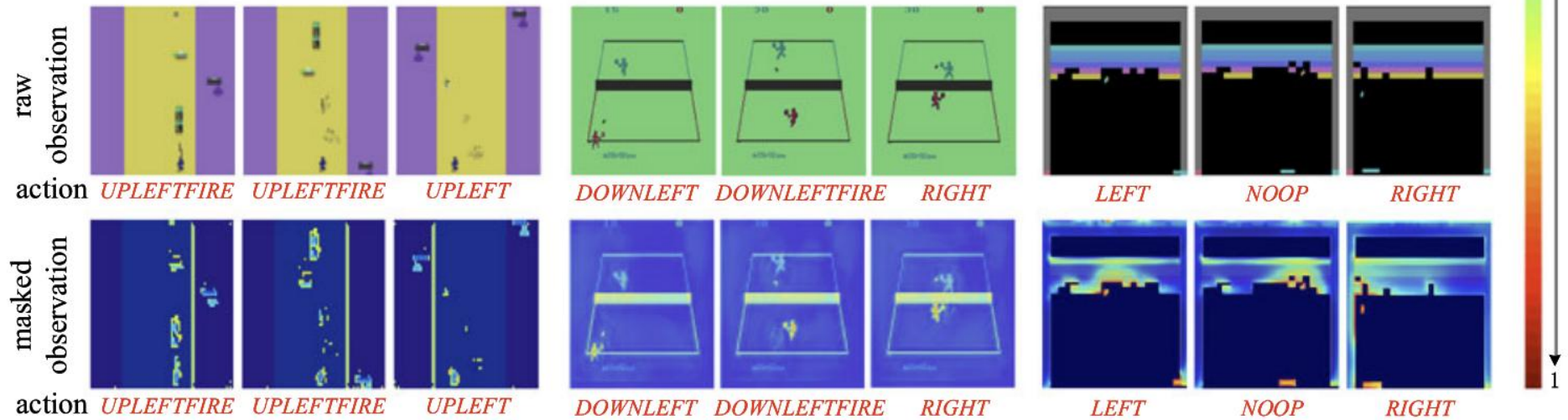




SSINet: The Output



(a) Lane-following

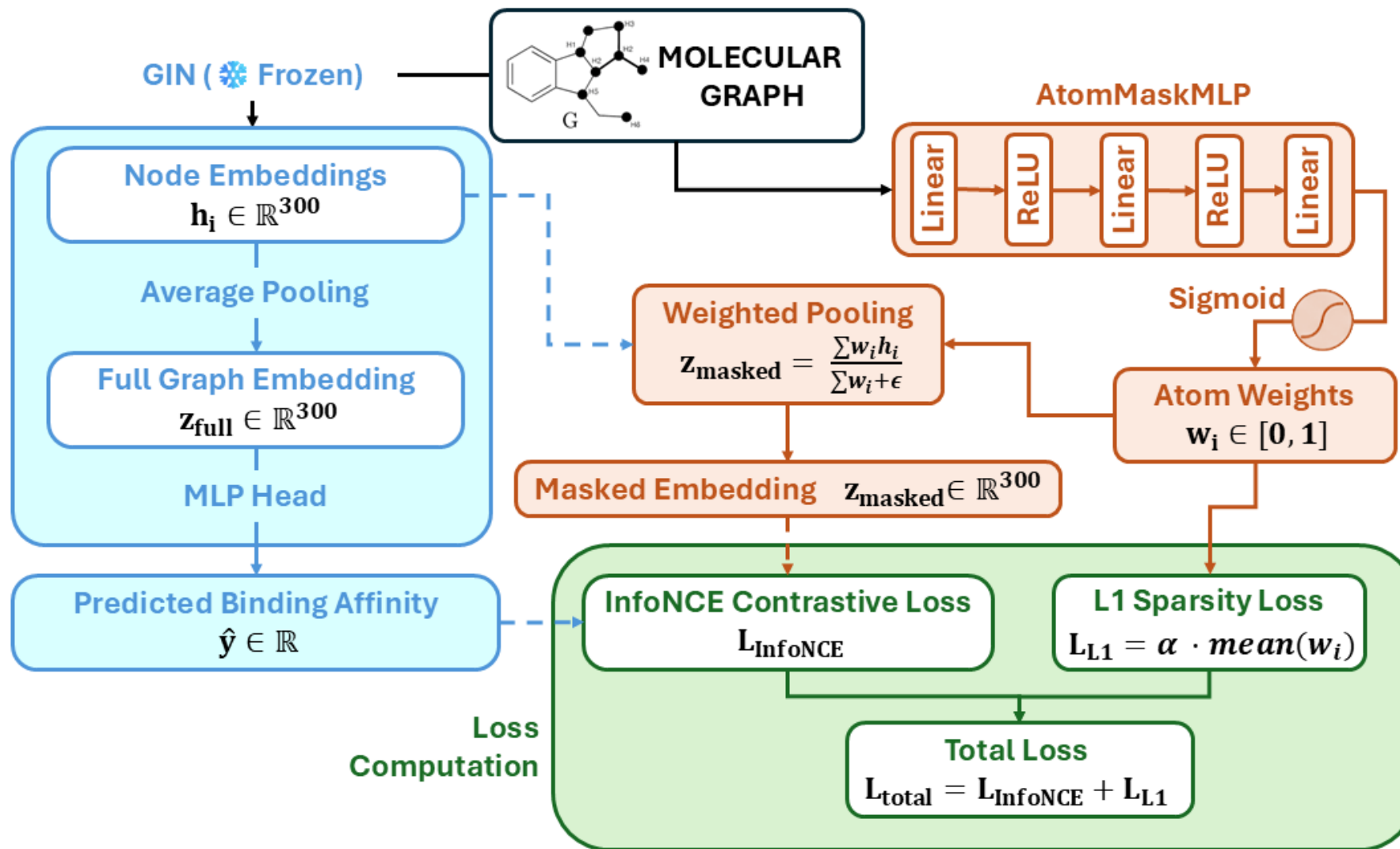


(b) Riverraid

(c) Tennis

(d) Breakout

The novelty: Molecular SSINet



The novelty: Molecular SSINet

$$L_{total} = \text{CrossEntropy}(S, \mathbf{y}) + \alpha \cdot \frac{1}{N} \sum_{i=1}^N w_i$$



Maximum Behavior Resemblance

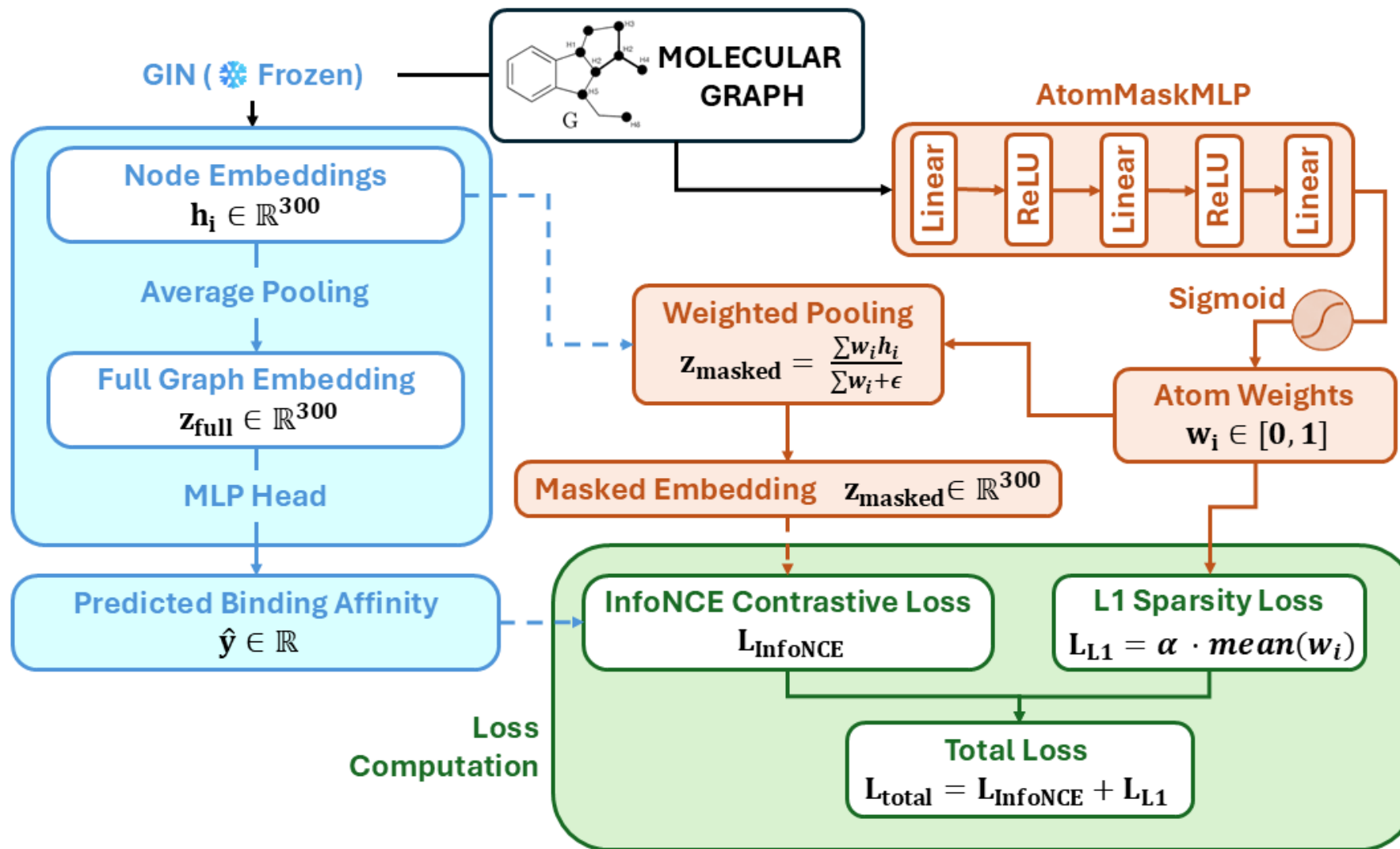
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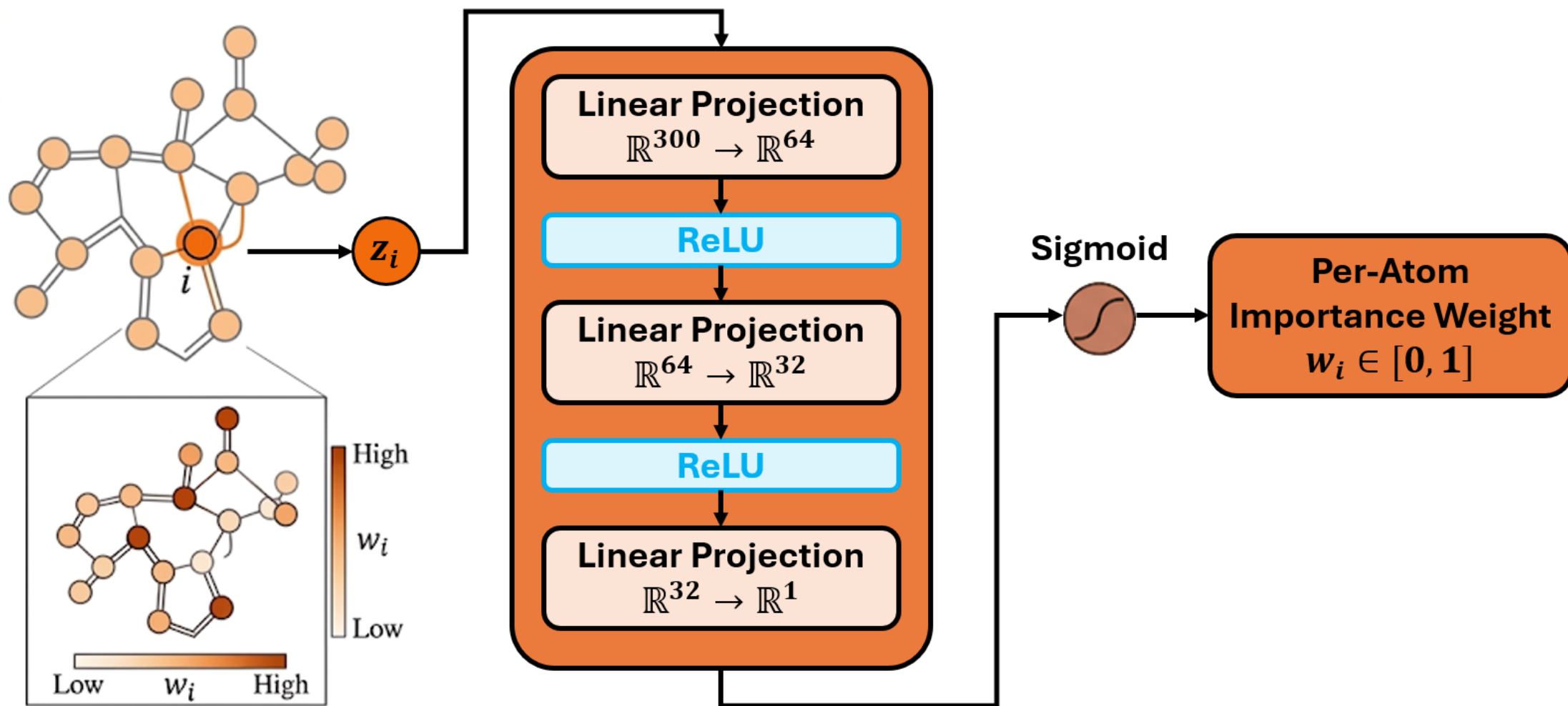
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The novelty: Molecular SSINet



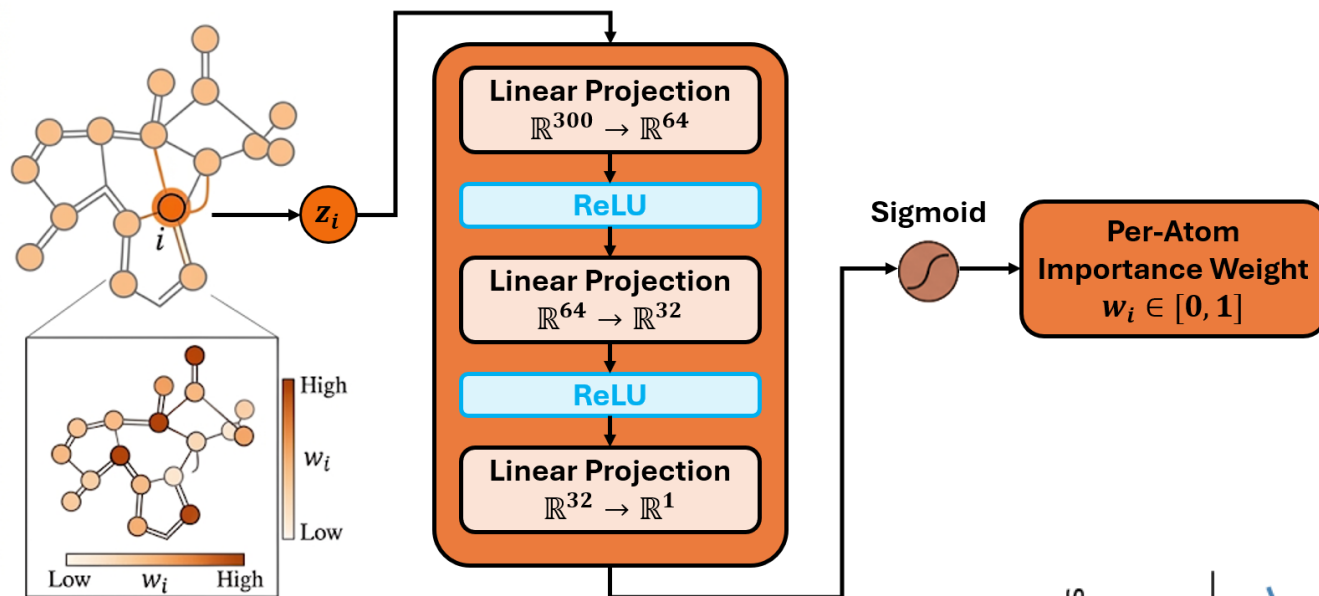


Molecular SSINet : Training The Mask



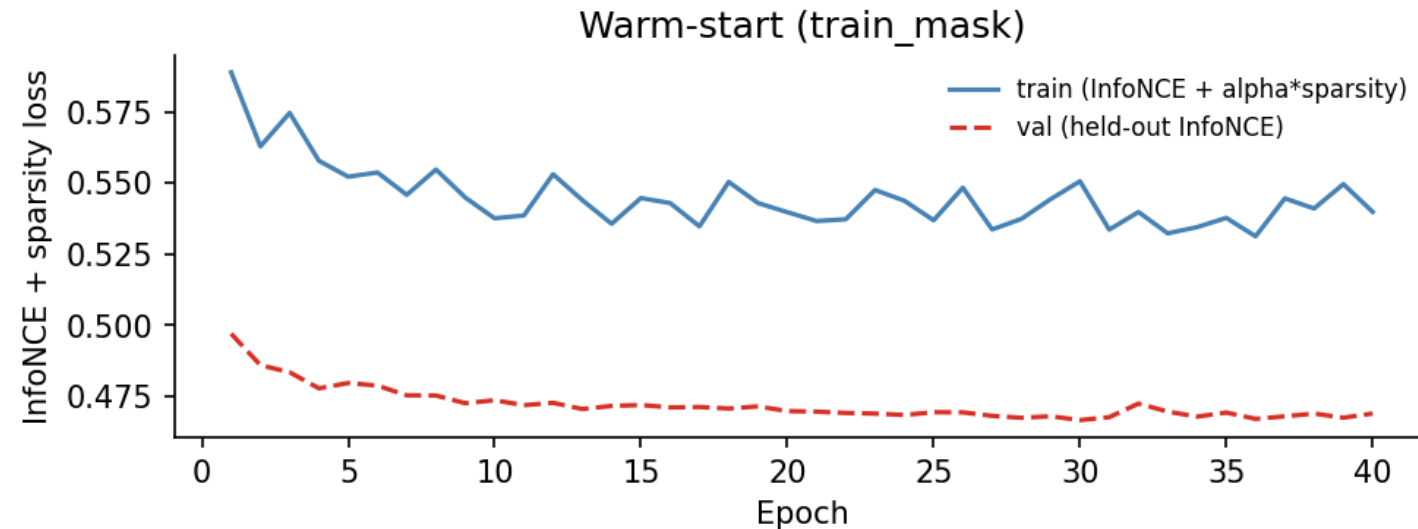


Molecular SSINet : Training The Mask



Warm-Start

First training on pretraining molecules to ensure **convergence**

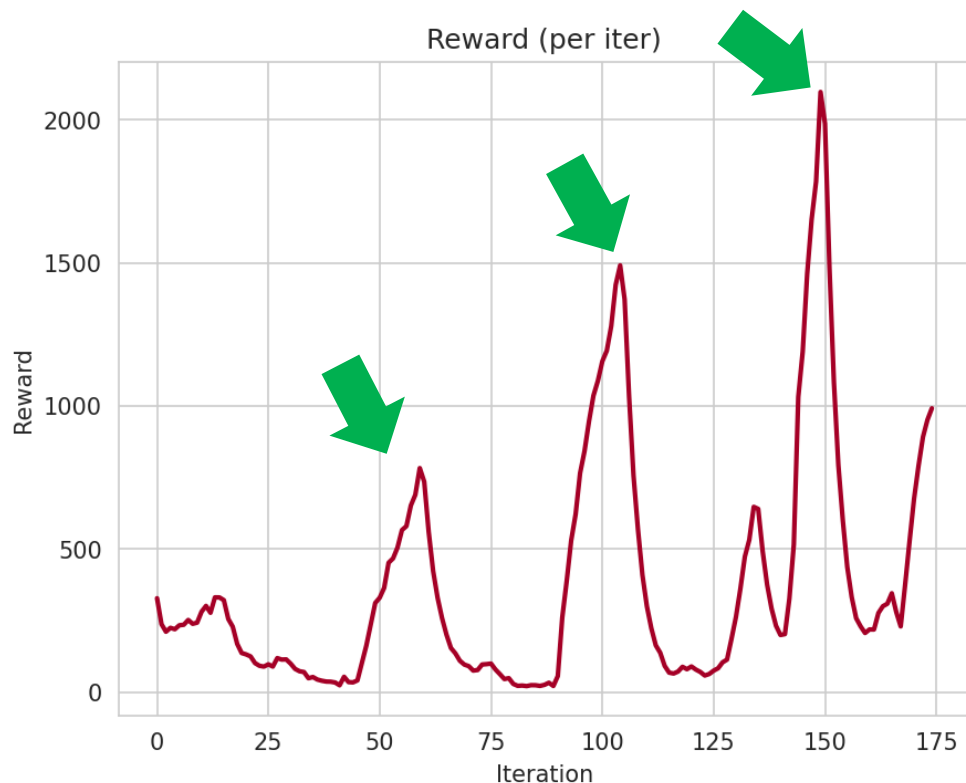




Molecular SSINet:



Training The Mask

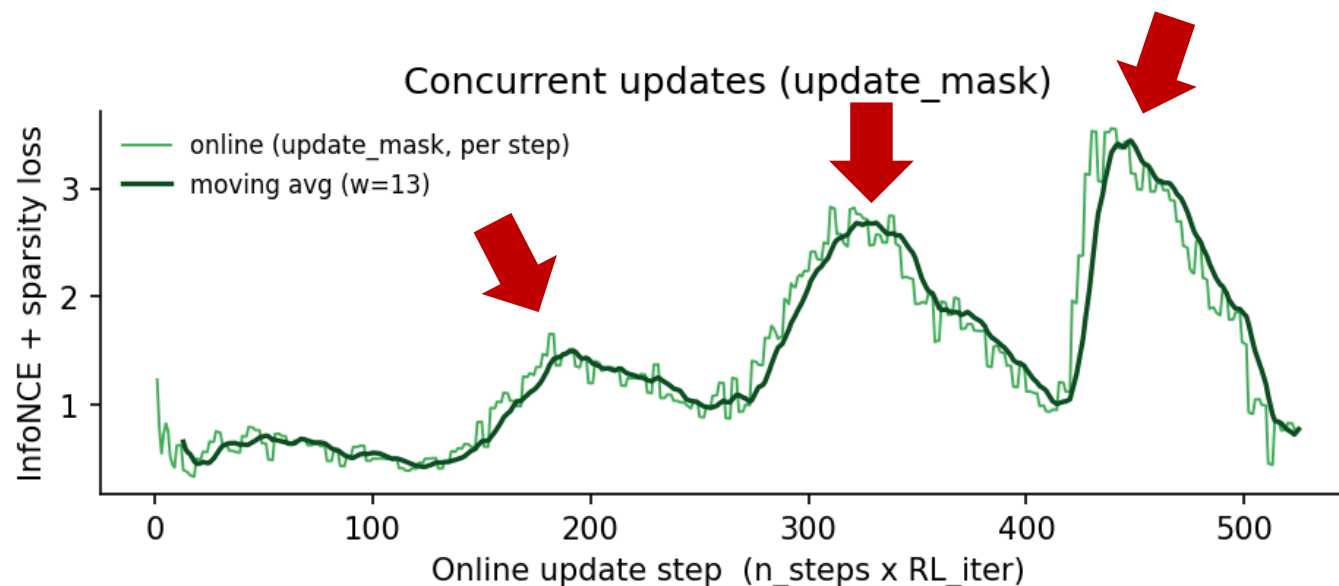


Same Trend: **Good Adaptation**

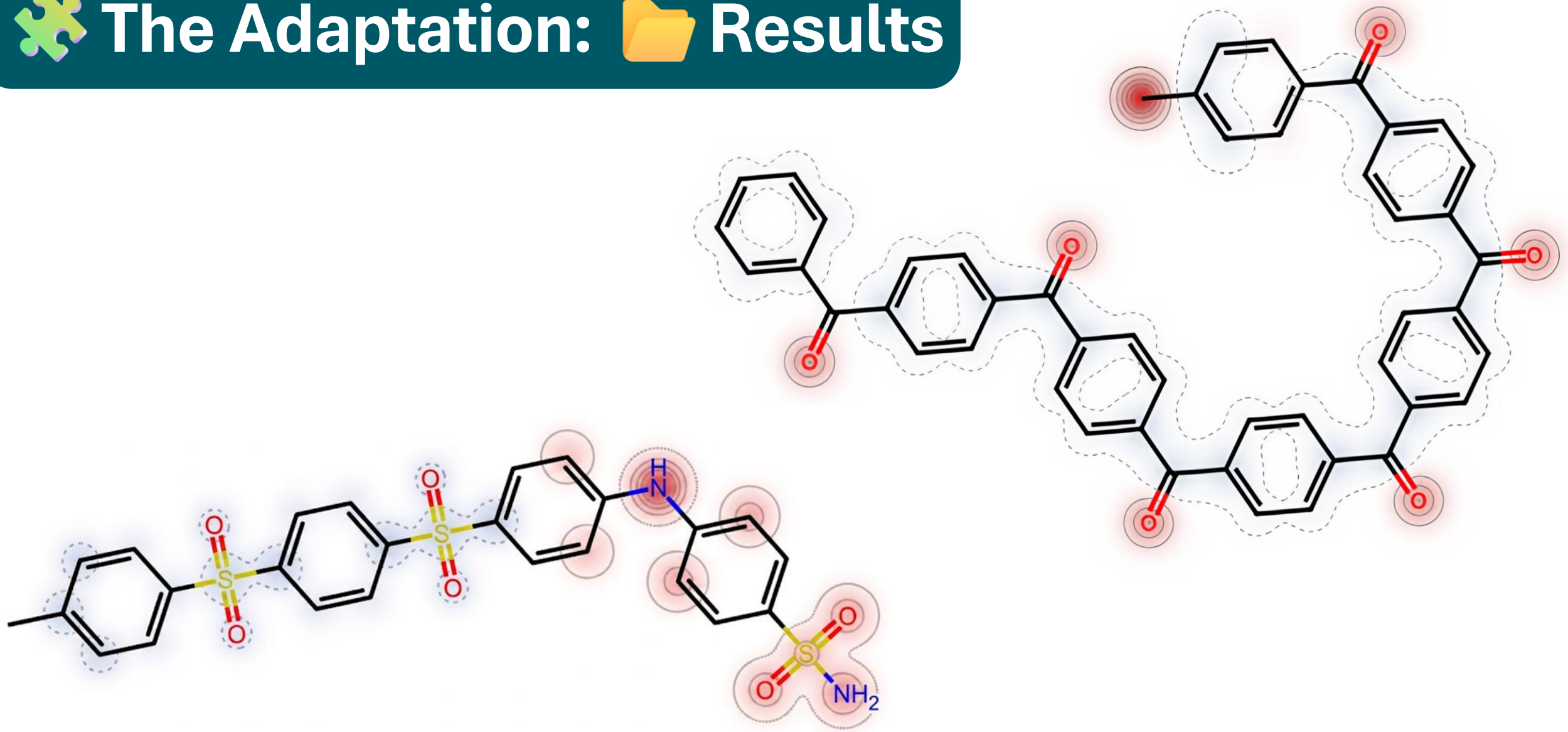
Quality is **training dependent**



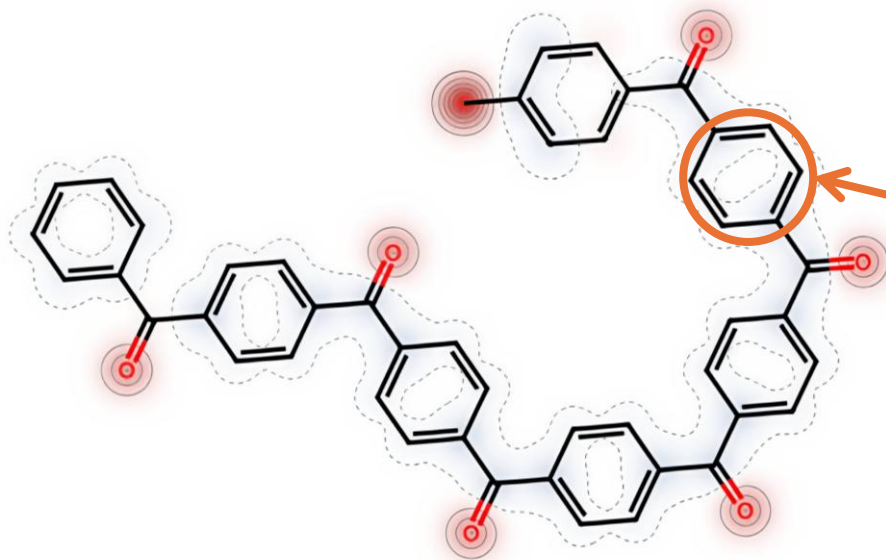

Concurrent Update
Three-steps-alignment after each generator iteration



The Adaptation: Results



The Adaptation: + Weighted Sum



Cc1ccc(C(=O)c2ccc(C(=O)c3ccc(C(=O)c4ccc(C(=O)c5ccc(C(=O)c6ccc(C(=O)c7c...))



The Mask is Confused

Having a lot of aromatic rings, it detect as **important anything that is different**



Reward Hacking

Greater **aromatic surface** area



More **hydrophobic** interactions



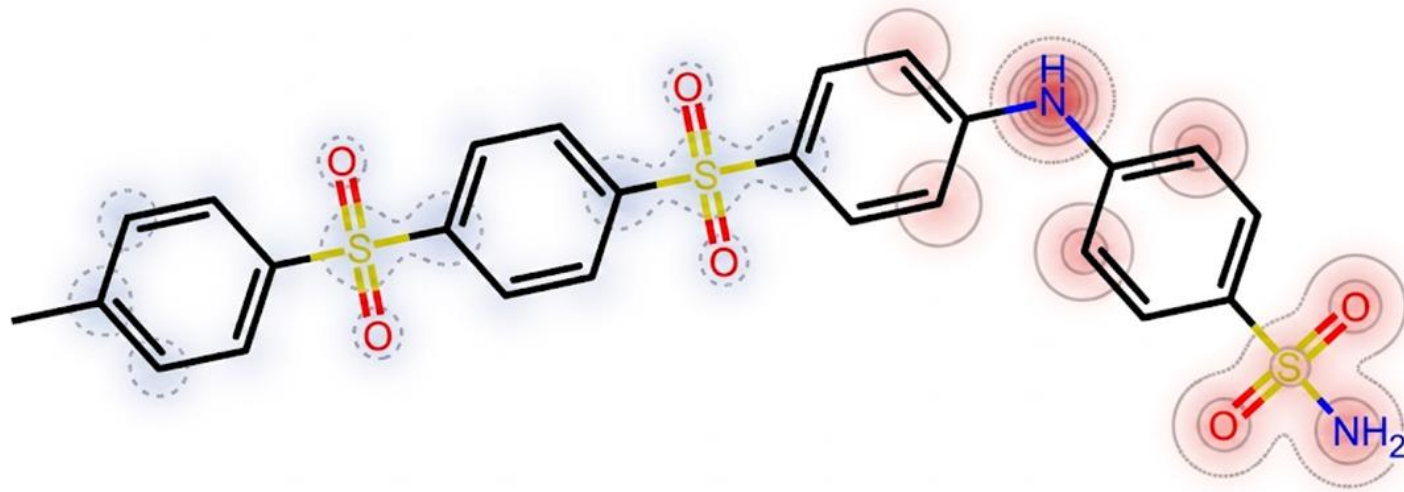
Higher **Binding Affinity**



Useless as a Drug

Does not have an **H-bond donor** to make **Hinge Contact** with the target protein,
still are detected important atoms

The Adaptation: Alternating Rewards (35)



Cc1ccc(S(=O)(=O)c2ccc(S(=O)(=O)c3ccc(Nc4ccc(S(N)(=O)=O)cc4)cc3)cc2)cc1



The Mask is Right!

Correctly detects the **N-H molecule** that breaks the previous malicious pattern.

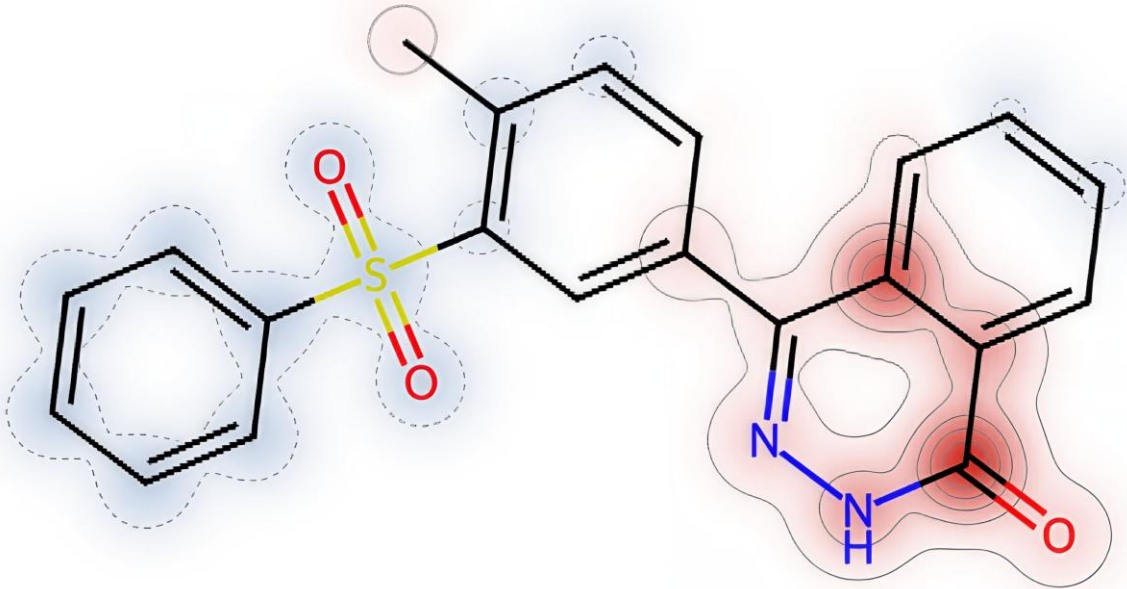


Linear oligomer of para-phenylene sulfone

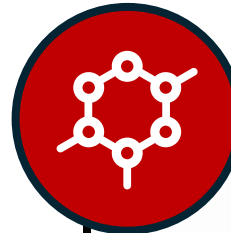
The alternating reward allowed the compound to incorporate a **primary sulfonamide group at the end**.

It is the responsible of the **hinge bond** that that **inhibits** the effects of the **target protein**.

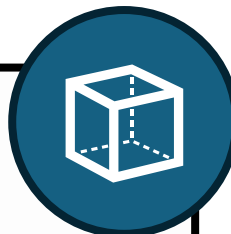
The Adaptation: Alternating Rewards (15)



Cc1ccc(C2=NNC(=O)c3ccccc32)cc1S(=O)(=O)c1ccccc1



The **phthalazinone fused-ring system**, acts as the core pharmacophore **providing the necessary H-bond** contacts to establish the **hinge binding** with TTBK1.



More structurally suitable and resistant than the **sulfonyle warhead** seen before, thanks to the presence of **Carbon** and **Nitrogen**.



Good **Adaptation**

Final Evaluation: Pros vs Cons



The mask demonstrated a good level of generalization



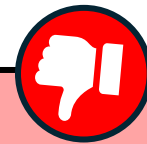
Around 3h of training with the chosen configuration



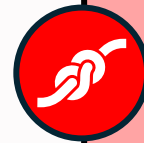
Good level of visual interpretability achieved for non-experts in the field



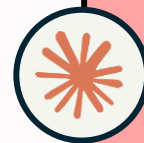
The properties of SSINet have been faithfully replicated



The quality of the mask is training dependent



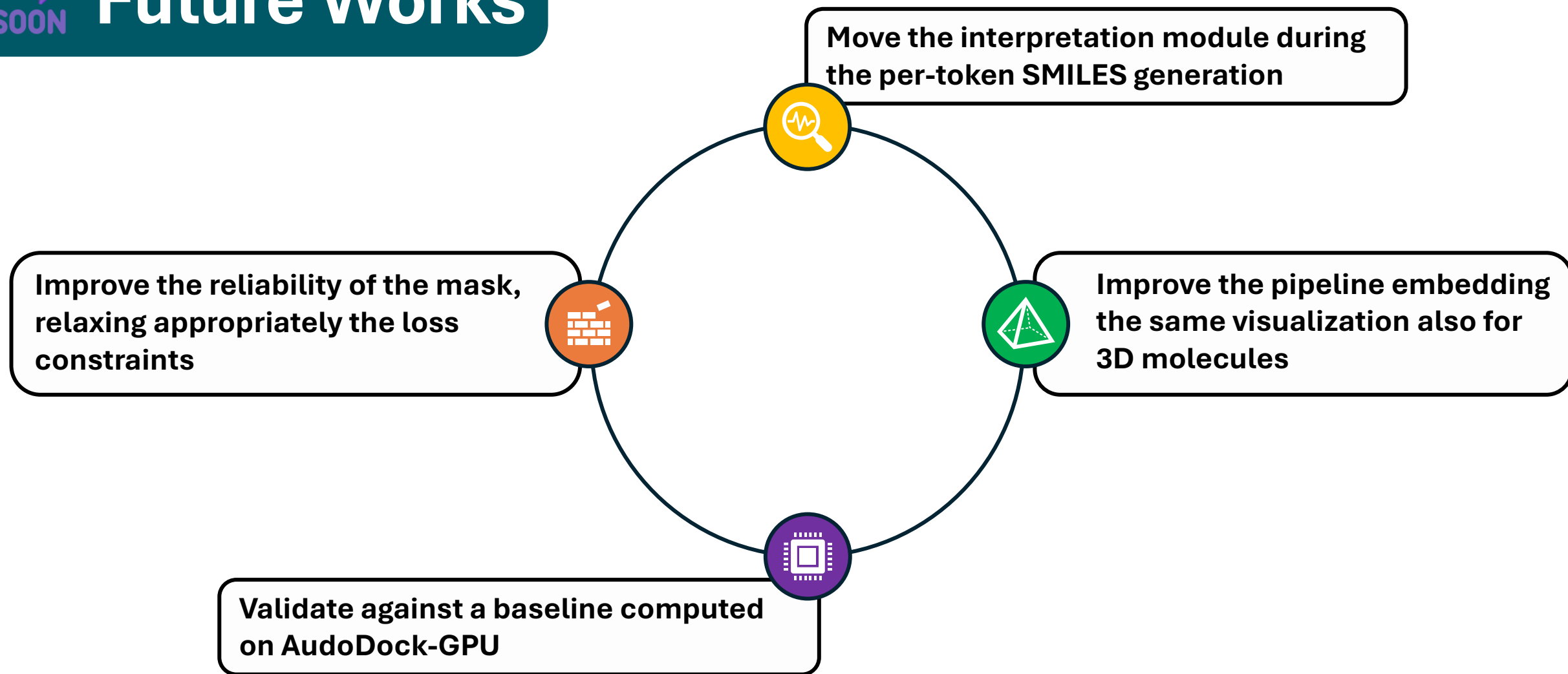
The importance is a per-molecule min-max norm, forcing every molecule to have important atoms



All the chemical evaluations were carried out using Claude Opus, instructed to act as a “Chemist in the Loop”



Future Works





**THANKS FOR YOUR
ATTENTION!**

Pimpinelli Francesco 2214340
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